

CPSC 304 Project Cover Page

Milestone #: 2

Date: February 22nd, 2026

Group Number: 83

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

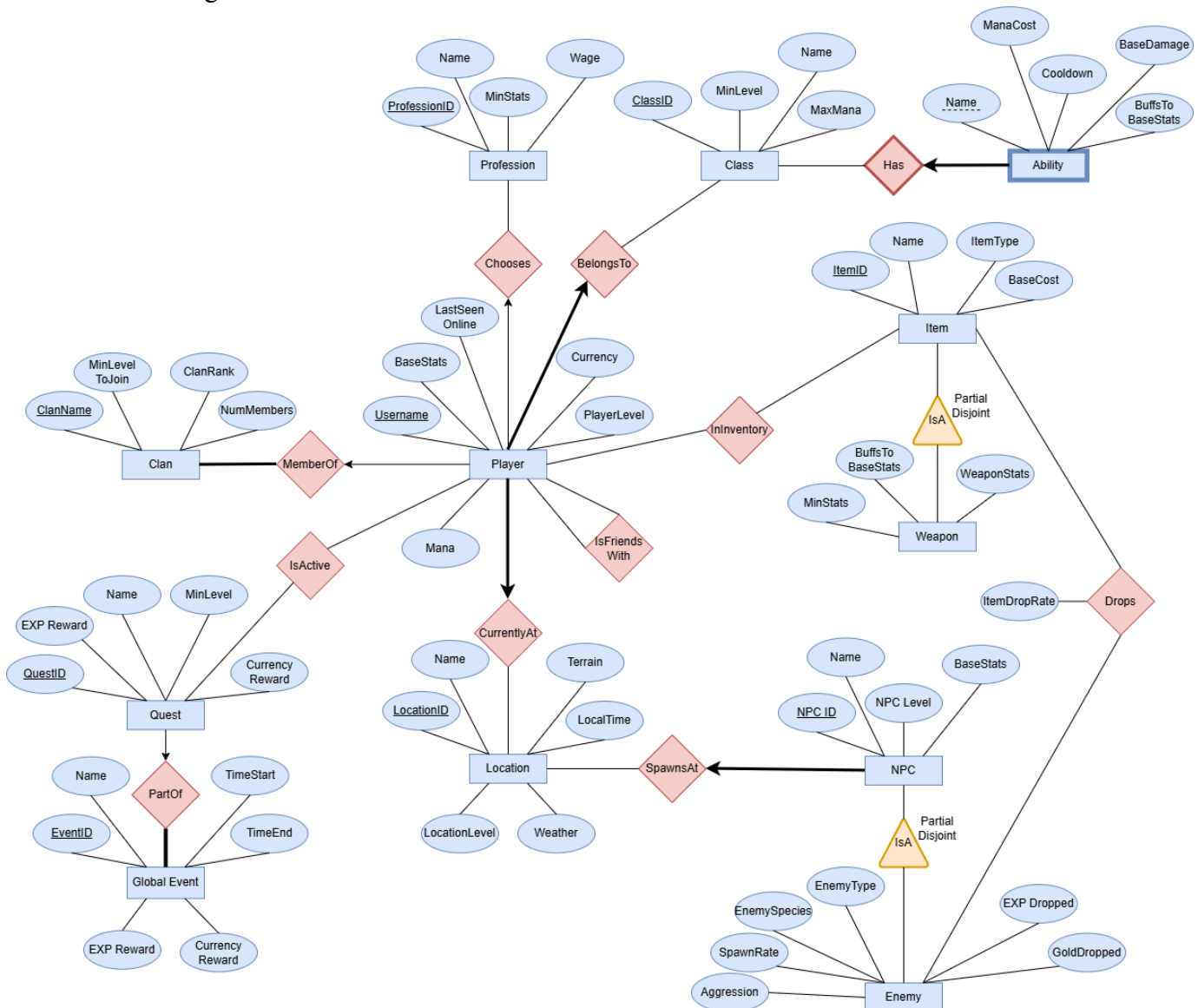
Summary of the project:

This project will model the management and moderation of a massively multiplayer online role-playing game (MMORPG). The database represents core game entities such as players, NPCs, items, quests, clans, locations, and much more.

Tech Stack:

We will be using Javascript/React with the department provided Oracle Database.

Final ER Diagram:



Between milestone 1 and milestone 2, the Level attribute of Player was renamed to PlayerLevel, the ItemStats attribute of Weapon was renamed to WeaponStats, and the EnemySpecies attribute was added to Enemy.

Schema:

1. Profession(ProfessionID: INT, Name: VARCHAR(50), Wage: INT, MinStats: VARCHAR(255))
PK: ProfessionID
CK: Name
Constraints: ProfessionID UNIQUE and NOT NULL, Name UNIQUE and NOT NULL
2. Class(ClassID: INT, Name: VARCHAR(50), MinLevel: INT, MaxMana: INT)
PK: ClassID
CK: Name
Constraints: ClassID UNIQUE and NOT NULL, Name UNIQUE and NOT NULL
3. Ability(Name: VARCHAR(50), ManaCost: INT, Cooldown: FLOAT(24), BaseDamage: INT, BuffsToBaseStats: VARCHAR(255), ClassID: INT)
PK: (ClassID, Name)
FK: ClassID references Class(ClassID)
Constraints: (ClassID, Name) UNIQUE and NOT NULL
4. Clan(ClanName: VARCHAR(25), MinLevelToJoin: INT, ClanRank: INT, NumMembers: INT)
PK: ClanName
Constraints: ClanName UNIQUE and NOT NULL
5. Player(Username: VARCHAR(25), PlayerLevel: INT, Currency: INT, Mana: INT, BaseStats: VARCHAR(255), LastSeenOnline: TIMESTAMP, ProfessionID: INT, ClassID: INT, ClanName: VARCHAR(25), LocationID: INT)
PK: Username
FK: ProfessionID references Profession(ProfessionID), ClassID references Class(ClassID), ClanName references Clan(ClanName), LocationID references Location(LocationID)
Constraints: Username UNIQUE and NOT NULL, ClassID NOT NULL, LocationID NOT NULL, PlayerLevel NOT NULL
6. IsFriendsWith(Player1: VARCHAR(25), Player2: VARCHAR(25))
PK: (Player1, Player2)
FK: Player1 references Player(Username), Player2 references Player(Username)
Constraints: (Player1, Player2) UNIQUE and NOT NULL

7. Item(ItemID: INT, Name: VARCHAR(50), ItemType: VARCHAR(50), BaseCost: INT)
PK: ItemID
Constraints: ItemID UNIQUE and NOT NULL
8. Weapon(ItemID: INT, WeaponStats: VARCHAR(255), MinStats: VARCHAR(255),
BuffsToBaseStats: VARCHAR(255))
PK: ItemID
FK: ItemID references Item(ItemID)
Constraints: ItemID UNIQUE and NOT NULL
9. InInventory(Username: VARCHAR(25), ItemID: INT)
PK: (Username, ItemID)
FK: Username references Player(Username), ItemID references Item(ItemID)
Constraints: (Username, ItemID) UNIQUE and NOT NULL
10. Location(LocationID: INT, Name: VARCHAR(50), Terrain: VARCHAR(50), LocalTime:
VARCHAR(50), LocationLevel: INT, Weather: VARCHAR(50))
PK: LocationID
Constraints: LocationID UNIQUE and NOT NULL
11. NPC(NPC_ID: INT, Name: VARCHAR(50), NPC_Level: INT, BaseStats:
VARCHAR(255), LocationID: INT)
PK: NPC_ID
FK: LocationID references Location(LocationID)
Constraints: NPC_ID UNIQUE and NOT NULL, LocationID NOT NULL
12. Enemy(NPC_ID: INT, EnemyType: VARCHAR(50), EnemySpecies: VARCHAR(50),
SpawnRate: FLOAT(24), Aggression: FLOAT(24), EXPDropped: INT, GoldDropped:
INT)
PK: NPC_ID
FK: NPC_ID references NPC(NPC_ID)
Constraints: NPC_ID UNIQUE and NOT NULL
13. Drops(NPC_ID: INT, ItemID: INT, ItemDropRate: FLOAT(24))
PK: (NPC_ID, ItemID)
FK: NPC_ID references Enemy(NPC_ID), ItemID references Item(ItemID)
Constraints: (NPC_ID, ItemID) UNIQUE and NOT NULL
14. Quest(QuestID: INT, Name: VARCHAR(50), MinLevel: INT, EXP_Reward: INT,
CurrencyReward: INT, EventID: INT)
PK: QuestID
FK: EventID references GlobalEvent(EventID)
Constraints: QuestID UNIQUE and NOT NULL

15. IsActive(Username: VARCHAR(25), QuestID: INT)

PK: (Username, QuestID)

FK: Username references Player(Username), QuestID references Quest(QuestID)

Constraints: (Username, QuestID) UNIQUE and NOT NULL

16. GlobalEvent(EventID: INT, Name: VARCHAR(50), TimeStart: TIMESTAMP, TimeEnd: TIMESTAMP, EXP_Reward: INT, CurrencyReward: INT)

PK: EventID

Constraints: EventID UNIQUE and NOT NULL

The participation constraint on Clan in the MemberOf relationship cannot be captured. The participation constraint on GlobalEvent in the PartOf relationship cannot be captured.

Part 6. Functional Dependencies (FDs):

- Player:
 - Username -> BaseStats, LastSeenOnline, Currency, PlayerLevel, Mana
 - Username -> ProfessionID
 - Username -> ClassID
 - Username -> LocationID
 - PlayerLevel, ClassID -> BaseStats
- Profession:
 - ProfessionID -> Name, MinStats, Wage
 - Name -> ProfessionID, MinStats, Wage
- Class:
 - ClassID -> Name, MinLevel, MaxMana
 - Name -> ClassID, MinLevel, MaxMana
- Ability:
 - ClassID, Name -> ManaCost, Cooldown, BaseDamage, BuffsToBaseStats
- Clan:
 - ClanName -> MinLevelToJoin, ClanRank, NumMembers
- Item:
 - ItemID -> Name, ItemType, BaseCost
- Weapon:
 - ItemID -> MinStats, BuffsToBaseStats, WeaponStats
- Location:
 - LocationID -> Name, Terrain, LocalTime, Weather, LocationLevel

- NPC:
NPC_ID -> Name, NPC_Level, BaseStats, LocationID
- Enemy:
NPC_ID -> EnemySpecies, ExpDropped, GoldDropped
EnemySpecies -> SpawnRate, Aggression, EnemyType
- Drops:
NPC_ID, ItemID -> ItemDropRate
- Quest:
QuestID -> Name, MinLevel, EXP_Reward, CurrencyReward, EventID
- GlobalEvent:
EventID -> Name, TimeStart, TimeEnd, EXP_Reward, CurrencyReward
EventID, TimeStart, TimeEnd -> EXP_Reward, CurrencyReward

Part 7. Normalised tables:

Trivial cases:

Profession(ProfessionID: INT, Name: VARCHAR(50), Wage: INT, MinStats: VARCHAR(255))

PK: ProfessionID

CK: Name

FK: none

Class(ClassID: INT, Name: VARCHAR(50), MinLevel: INT, MaxMana: INT)

PK: ClassID

CK: Name

FK: none

Ability(Name: VARCHAR(50), ManaCost: INT, Cooldown: FLOAT(24), BaseDamage: INT, BuffsToBaseStats: VARCHAR(255), ClassID: INT)

PK: (ClassID, Name)

CK: N/A

FK: ClassID references Class(ClassID)

Clan(ClanName: VARCHAR(25), MinLevelToJoin: INT, ClanRank: INT, NumMembers: INT)

PK: ClanName

CK: N/A

FK: none

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IsFriendsWith(Player1: VARCHAR(25), Player2: VARCHAR(25))

PK: (Player1, Player2)

CK: (Player1, Player2)

FK: Player1 references Player(Username), Player2 references Player(Username)

Item(ItemID: INT, Name: VARCHAR(50), ItemType: VARCHAR(50), BaseCost: INT)

PK: ItemID

CK: N/A

FK: none

Weapon(ItemID: INT, WeaponStats: VARCHAR(255), MinStats: VARCHAR(255),

BufsToBaseStats: VARCHAR(255))

PK: ItemID

CK: N/A

FK: ItemID references Item(ItemID)

InInventory(Username: VARCHAR(25), ItemID: INT)

PK: (Username, ItemID)

CK: N/A

FK: Username references Player(Username), ItemID references Item(ItemID)

Location(LocationID: INT, Name: VARCHAR(50), Terrain: VARCHAR(50), LocalTime:

VARCHAR(50), LocationLevel: INT, Weather: VARCHAR(50))

PK: LocationID

CK: N/A

FK: N/A

NPC(NPC_ID: INT, Name: VARCHAR(50), NPC_Level: INT, BaseStats: VARCHAR(255),

LocationID: INT)

PK: NPC_ID

CK: N/A

FK: LocationID references Location(LocationID)

Drops(NPC_ID: INT, ItemID: INT, ItemDropRate: FLOAT(24))

PK: (NPC_ID, ItemID)

CK: N/A

FK: NPC_ID references Enemy(NPC_ID), ItemID references Item(ItemID)

Quest(QuestID: INT, Name: VARCHAR(50), MinLevel: INT, EXP_Reward: INT,

CurrencyReward: INT, EventID: INT)

PK: QuestID

CK: N/A

FK: EventID references GlobalEvent(EventID)

IsActive(Username: VARCHAR(25), QuestID: INT)

PK: (Username, QuestID)

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CK: N/A

FK: Username references Player(Username), QuestID references Quest(QuestID)

GlobalEvent(EventID: INT, Name: VARCHAR(50), TimeStart: TIMESTAMP, TimeEnd: TIMESTAMP, EXP_Reward: INT, CurrencyReward: INT)

PK: EventID

CK: N/A

FK: N/A

Violating tables:

Enemy table breaks BCNF since, by definition of BCNF every non-trivial FD should have the determinant being the superkey, however, EnemySpecies is *not* a superkey. Normalising Enemy requires decomposing Enemy on the FD EnemySpecies → SpawnRate, Aggression, EnemyType:

Enemy(NPC_ID: INT, EnemySpecies: VARCHAR(50), EXPDropped: INT, GoldDropped: INT)

PK: NPC_ID

CK: N/A

FK: NPC_ID references NPC(NPC_ID), EnemySpecies references EnemySpecies(EnemySpecies)

EnemySpecies(EnemySpecies: VARCHAR(50), SpawnRate: FLOAT(24), Aggression: FLOAT(24), EnemyType: VARCHAR(50))

PK: EnemySpecies

CK: N/A

FK: N/A

Similarly, Player table breaks BCNF and requires decomposition into two separate tables, since PlayerLevel is not a superkey:

Player(Username: VARCHAR(25), PlayerLevel: INT, Currency: INT, Mana: INT, LastSeenOnline: TIMESTAMP, ProfessionID: INT, ClanName: VARCHAR(25), ClassID: INT, LocationID: INT)

PK: Username

CK: N/A

FK: ProfessionID references Profession(ProfessionID), ClanName references Clan(ClanName), LocationID references Location(LocationID), (PlayerLevel, ClassID) references PlayerLevel(BaseStats)

PlayerLevel(PlayerLevel: INT, ClassID: INT, BaseStats: VARCHAR(255))

PK: (PlayerLevel, ClassID)

CK: N/A

FK: ClassID references Class(ClassID)

SQL DDL:

All of the foreign keys should have “ON UPDATE CASCADE”, but since Oracle does not support ON UPDATE, all foreign keys will instead use the default update behaviour (equivalent to ON UPDATE NO ACTION).

The SQL DDL statements required to create all the normalised tables are as follows:

- CREATE TABLE Profession(
 ProfessionID INT,
 Name VARCHAR(50) NOT NULL UNIQUE,
 Wage INT,
 MinStats VARCHAR(255),
 PRIMARY KEY(ProfessionID)
);
- CREATE TABLE Class(
 ClassID INT,
 Name VARCHAR(50) NOT NULL UNIQUE,
 MinLevel INT,
 MaxMana INT,
 PRIMARY KEY(ClassID)
);
- CREATE TABLE Ability(
 ClassID INT,
 Name VARCHAR(50),
 ManaCost INT,
 Cooldown FLOAT(24),
 BaseDamage INT,
 BuffsToBaseStats VARCHAR(255),
 PRIMARY KEY(ClassID, Name),
 FOREIGN KEY(ClassID) references Class ON DELETE CASCADE
);
- CREATE TABLE Clan(
 ClanName VARCHAR(25),
 MinLevelToJoin INT,
 ClanRank INT,
 NumMembers INT,
 PRIMARY KEY(ClanName)
);

- CREATE TABLE GlobalEvent(
 EventID INT,
 Name VARCHAR(50),
 TimeStart TIMESTAMP,
 TimeEnd TIMESTAMP,
 EXP_Reward INT,
 CurrencyReward INT,
 PRIMARY KEY(EventID)
);
- CREATE TABLE Quest(
 QuestID INT,
 Name VARCHAR(50),
 MinLevel INT,
 EXP_Reward INT,
 CurrencyReward INT,
 EventID INT,
 PRIMARY KEY(QuestID),
 FOREIGN KEY(EventID) references GlobalEvent ON DELETE SET NULL
);
- CREATE TABLE Location(
 LocationID INT,
 Name VARCHAR(50),
 Terrain VARCHAR(50),
 LocalTime VARCHAR(50),
 LocationLevel INT,
 Weather VARCHAR(50),
 PRIMARY KEY(LocationID)
);
- CREATE TABLE NPC(
 NPC_ID INT,
 Name VARCHAR(50),
 NPC_Level INT,
 BaseStats VARCHAR(255),
 LocationID INT NOT NULL,
 PRIMARY KEY(NPC_ID),
 FOREIGN KEY(LocationID) references Location ON DELETE CASCADE
);
- CREATE TABLE EnemySpecies(
 EnemySpecies VARCHAR(50),
 SpawnRate FLOAT(24),
 Aggression FLOAT(24),
 EnemyType VARCHAR(50),

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        PRIMARY KEY(EnemySpecies)
    );

- CREATE TABLE Enemy(
    NPC_ID INT,
    EnemySpecies VARCHAR(50) NOT NULL,
    EXPDropped INT,
    GoldDropped INT,
    PRIMARY KEY(NPC_ID),
    FOREIGN KEY(NPC_ID) references NPC ON DELETE CASCADE
    FOREIGN KEY(EnemySpecies) references EnemySpecies ON DELETE
    CASCADE
);

- CREATE TABLE Item(
    ItemID INT,
    Name VARCHAR(50),
    ItemType VARCHAR(50),
    BaseCost INT,
    PRIMARY KEY(ItemID)
);

- CREATE TABLE Weapon(
    ItemID INT,
    WeaponStats VARCHAR(255),
    MinStats VARCHAR(255),
    BuffsToBaseStats VARCHAR(255),
    PRIMARY KEY(ItemID),
    FOREIGN KEY(ItemID) references Item ON DELETE CASCADE
);

- CREATE TABLE Drops(
    NPC_ID INT,
    ItemID INT,
    ItemDropRate FLOAT(24),
    PRIMARY KEY(NPC_ID, ItemID),
    FOREIGN KEY(NPC_ID) references Enemy ON DELETE CASCADE,
    FOREIGN KEY(ItemID) references Item ON DELETE CASCADE
);

- CREATE TABLE PlayerLevel(
    PlayerLevel INT,
    ClassID INT NOT NULL,
    BaseStats VARCHAR(255),
    PRIMARY KEY(PlayerLevel, ClassID),
    FOREIGN KEY(ClassID) references Class
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);

- CREATE TABLE Player(
    Username VARCHAR(25),
    PlayerLevel INT NOT NULL,
    Currency INT,
    Mana INT,
    LastSeenOnline TIMESTAMP,
    ProfessionID INT,
    ClassID INT NOT NULL,
    ClanName VARCHAR(25),
    LocationID INT NOT NULL,
    PRIMARY KEY(Username),
    FOREIGN KEY(ProfessionID) references Profession ON DELETE SET NULL,
    FOREIGN KEY(ClassID) references Class,
    FOREIGN KEY(ClanName) references Clan ON DELETE SET NULL,
    FOREIGN KEY(LocationID) references Location,
    FOREIGN KEY(PlayerLevel, ClassID) references PlayerLevel(PlayerLevel,
ClassID)
);

- CREATE TABLE InInventory(
    Username VARCHAR(25),
    ItemID INT,
    PRIMARY KEY(Username, ItemID),
    FOREIGN KEY(Username) references Player ON DELETE CASCADE,
    FOREIGN KEY(ItemID) references Item ON DELETE CASCADE
);

- CREATE TABLE IsActive(
    Username VARCHAR(25),
    QuestID INT,
    PRIMARY KEY(Username, QuestID),
    FOREIGN KEY(Username) references Player ON DELETE CASCADE,
    FOREIGN KEY(QuestID) references Quest ON DELETE CASCADE
);

- CREATE TABLE IsFriendsWith(
    Player1 VARCHAR(25),
    Player2 VARCHAR(25),
    PRIMARY KEY(Player1, Player2),
    FOREIGN KEY(Player1) references Player ON DELETE CASCADE,
    FOREIGN KEY(Player2) references Player ON DELETE CASCADE
);
```

AI declaration: No AI tools have been used in the process of completing Milestone 2.